


# Report

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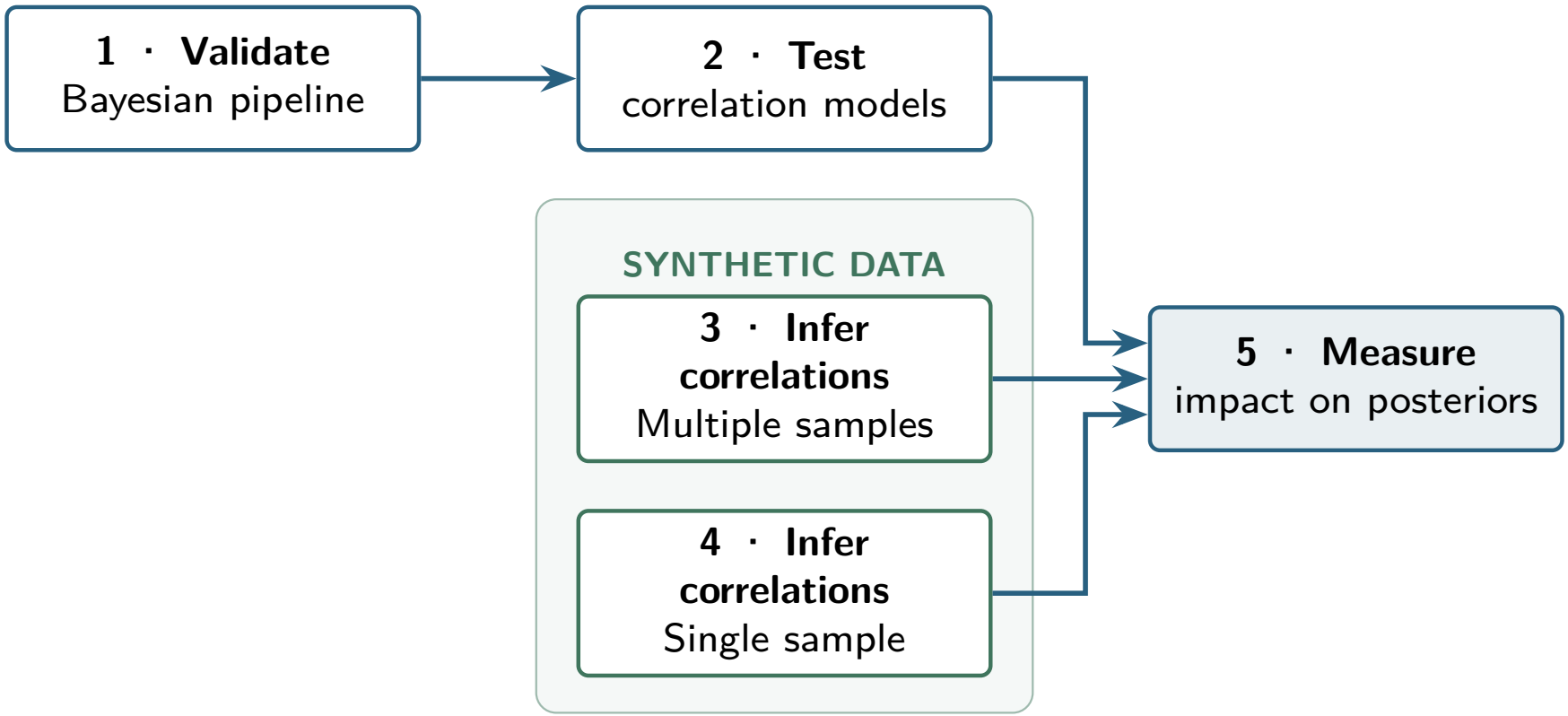
# Summary

1. Tasks Status
2. Selected Article

# Conference Materials Status

| Task | Status | |
|---|---|---|
| **2026-10 Conference** | ○ | Simpósio IFUSP Presentation |
| **2026-10 Conference** | ○ | AGORÁ Presentation |
| **2026-11 Poster** | ○ | 3rd Schooll QC Poster |
| **2026-12 Conference** | ○ | LASNPA Presentation |
| **2026-12 Paper** | ○ | LASNPA Proceedings |
| **2027-03 Conference** | ○ | Quark Matter Presentation |

# Study Phases – Dependencies



# Phase 1 – Bayesian Pipeline Validation

| **Task** | **Status** | **Output** |
|---|---|---|
| ↪ Reproduce Mankolli et al. results with the original JETSCAPE emulator | ✓ | Reference |
| ↪ Reproduce the same results with the `hic_bayes` emulator | ✓ | Validation |
| ↪ Emulator closure test | ✓ | Validation |
| ↪ MCMC convergence test | ✓ | Diagnostics |

# Phase 2 – Correlation Model Sensitivity

| Task | Status | Output |
|---|---|---|
| ↪ Au–Au-200 + d–Au-200: Correlation-model sensitivity | | Posteriors |
| • Global CS correlation | ✓ | Notebook |
| • Global RBF correlation | ✓ | Notebook |
| • Global RBF correlation fix unit | ◔ | Notebook |
| • Detector-grouped CS correlation | ◑ | Notebook |
| • Detector-grouped RBF correlation | ◑ | Notebook |
| ↪ Compare posterior shifts, credible intervals, and parameter correlations | | Analysis |
| • Global correlation models | ✓ | Notebook |
| • Detector-grouped correlation models | ◑ | Notebook |
| ↪ Quantify and compare the emulator-error and experimental-error covariance matrices in the likelihood | ◔ | Analysis |

# Phase 3 – Correlation Inference from Multiple Samples

| **Task** | **Status** | **Output** |
|---|---|---|
| ↪ Generate controlled synthetic datasets with known covariance ($N$ samples; goal: $N = 10$) | ◔ | Dataset |
| ↪ Classical correlation inference | ○ | Benchmark |
| ↪ Quantum-learning correlation inference | ○ | Benchmark |
| ↪ Classical vs. quantum comparison against the known covariance | ○ | Comparison |

# Phase 4 – Single-Sample Correlation Inference

| **Task** | **Status** | **Output** |
|---|---|---|
| ↪ Study correlation approximations when only one experimental sample is available | ◔ | Heuristics |
| ↪ Use reported uncertainty components to constrain plausible correlation structures | ○ | Models |

# Phase 5 – End-to-End Bayesian Impact

| **Task** | **Status** | **Output** |
|---|---|---|
| ↪ Propagate inferred covariance matrices through the Bayesian likelihood<br>• **Single sample:** CS, RBF, Zero, GP<br>• **Multiple samples:** Classical, Quantum | ◔ | Posteriors |
| ↪ Quantify covariance-estimation error → posterior error | ○ | Final study |

# Selected Article

**Thermodynamic and Transport Properties of Quark-Gluon Plasma at Finite Chemical Potential with a DNN framework**

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(Dated: April 8, 2026)

The characteristics of a thermal system depend strongly on its response to thermal gradients and the underlying microscopic interactions among constituents. In the present study, we investigate the thermodynamic and transport properties of the quark-gluon plasma (QGP) at finite baryon chemical potential within a deep-learning-assisted quasi-particle model (DLQPM). The temperature (T) and baryon chemical potential ($\mu_B$)-dependent thermal masses of quasi-particles are estimated using neural networks trained to reproduce lattice QCD (lQCD) results for the equation of state, obtained via a Taylor-like expansion around vanishing baryon chemical potential. The trained model acts as an effective emulator, enabling us to estimate the thermodynamic and transport properties at finite $\mu_B$. We compute the speed of sound, specific heat, viscosity, and conductivity of the deconfined medium. Our findings are in good agreement with available lattice calculations and other phenomenological models. The present study demonstrates that a DNN-based approach provides an efficient framework for studying the properties of the QGP at finite baryon density.

Figure: Alternative methods to infer transport parameters of QGP